

Q3. Importance of Test Cases & Test Plans

Scenario: Inventory Management System | **Modules:** Add Item, Update Stock, Generate Reports

1. Introduction

In software quality assurance, **test cases** and **test plans** are two foundational artifacts that work together to ensure a system functions correctly, reliably, and as expected by stakeholders. For an Inventory Management System (IMS) with modules for Add Item, Update Stock, and Generate Reports, both artifacts are essential to prevent defects, validate business logic, and maintain system integrity.

2. What Are Test Cases?

A **test case** is a documented set of conditions, inputs, actions, and expected outcomes used to verify that a specific feature or function behaves correctly. Each test case targets one particular behaviour of the system.

Key Components of a Test Case:

Component	Description	Example (Update Stock Module)
Test Case ID	Unique identifier	TC-US-01
Test Title	Short description of what is being tested	Update stock with valid quantity
Preconditions	State required before test execution	User ITM001 exists and is active
Input Data	Values/data entered into the system	Item ID: ITM001, Quantity: 150
Test Steps	Step-by-step actions performed	1. Open Update Stock 2. Enter ID 3. Enter qty 4. Submit
Expected Result	What should happen if system works correctly	Stock for ITM001 updated to 150; success message shown
Actual Result	What actually happened during execution	File status after test run
Status	Pass / Fail / Blocked	PASS

Why Test Cases Are Important:

- **Defect Detection:** They expose bugs before the system reaches production.
- **Requirement Verification:** Each test case maps back to a functional requirement, ensuring nothing is missed.
- **Repeatability:** The same test can be re-run after changes to confirm no regression.
- **Documentation:** They serve as evidence of testing for audits and compliance.
- **Coverage:** They ensure all paths — valid, invalid, and edge cases — are tested.

3. Sample Test Cases — Inventory Management System

Module 1: Add Item

TC ID	Test Scenario	Input	Expected Result	Type
TC-AI-01	Add item with all valid fields	Name: Pen, Qty: 100, Price: 5.00	Item added; confirmation shown	Positive
TC-AI-02	Add item with duplicate item ID	ID: ITM001 (exists)	ERROR: Item already exists	Negative
TC-AI-03	Add item with blank name field	Name: " (empty)	ERROR: Name is required	Negative

TC-AI-04	Add item with quantity = 0	Qty: 0	Item added with zero stock	Boundary
TC-AI-05	Add item with negative price	Price: -10	ERROR: Price must be positive	Negative

Module 2: Update Stock

TC ID	Test Scenario	Input	Expected Result	Type
TC-US-01	Update stock for valid active item	ID: ITM001, Qty: 200	Stock updated to 200	Positive
TC-US-02	Update stock to zero	ID: ITM001, Qty: 0	Stock set to 0; warning shown	Boundary
TC-US-03	Update stock — item not found	ID: UNKNOWN, Qty: 50	ERROR: Item not found	Negative
TC-US-04	Update stock with negative quantity	ID: ITM001, Qty: -5	ERROR: Quantity cannot be negative	Negative
TC-US-05	Update stock exceeding maximum	ID: ITM001, Qty: 10001	ERROR: Exceeds maximum stock	Boundary

Module 3: Generate Reports

TC ID	Test Scenario	Input	Expected Result	Type
TC-GR-01	Generate report with valid date range	From: 01/06 To: 30/06	Report lists all items in range	Positive
TC-GR-02	Generate report — no data in range	From: 01/01 To: 01/01	Report shows 'No records found'	Negative
TC-GR-03	Generate report for 1000+ items	All items selected	Report generated within 5 seconds	Performance
TC-GR-04	Report data matches actual stock	All values	Values in report = values in DB	Integration
TC-GR-05	Export report to PDF/Excel	Format: PDF	File downloaded successfully	Functional

4. What Is a Test Plan?

A **test plan** is a comprehensive document that describes the overall testing strategy for a system or project. It defines the *scope*, *approach*, *resources*, *schedule*, and *criteria* for testing. While a test case answers '*what to test*,' the test plan answers '*how, when, who, and why*.'

Key Sections of a Test Plan:

Section	Purpose	IMS Example
1. Objective	State what testing aims to achieve	Verify all 3 IMS modules meet functional requirements
2. Scope	Define what is included / excluded	In scope: Add Item, Update Stock, Reports. Out: Login module
3. Test Strategy	Approach — types of testing to be performed	Functional, boundary, negative, integration, performance tests
4. Test Environment	Hardware, software, tools required	Windows 10, Chrome 120, MySQL 8.0, Selenium, JIRA
5. Test Schedule	Timeline and milestones	Unit testing: Week 1; Integration: Week 2; UAT: Week 3
6. Roles & Responsibilities	Who does what	QA Lead: plan & review; Testers: execute; Dev: fix defects
7. Entry Criteria	Conditions before testing can begin	Build deployed to test env; requirements document approved
8. Exit Criteria	Conditions required to conclude testing	95% test cases passed; no critical defects open
9. Risk & Mitigation	Potential risks and how to handle them	Risk: bulk report timeout — Mitigation: performance testing early
10. Deliverables	Documents/reports produced by QA	Test cases, defect reports, test summary report

Why Test Plans Are Important:

- **Organisation:** Ensures all team members understand their testing responsibilities and timelines.
- **Completeness:** Prevents critical modules from being overlooked — scope is explicitly defined.
- **Risk Management:** Identifies high-risk areas early (e.g., bulk report generation) so they are prioritised.
- **Traceability:** Links test activities to requirements, providing proof of coverage.
- **Stakeholder Confidence:** Gives management visibility into QA progress and quality gates.

5. Relationship Between Test Cases and Test Plans

Aspect	Test Plan	Test Case
Definition	Strategic document for entire testing	Specific check for one feature/behaviour
Scope	Covers whole system or module	Covers a single scenario
Level	High-level	Detailed / granular
Created by	QA Lead / Test Manager	QA Engineer / Tester
Timing	Before testing begins	During test design phase
IMS Example	Plan covering all 3 modules	TC-US-03: Update stock for non-existent item
Purpose	Defines how testing will be managed	Validates one specific behaviour

6. Conclusion

Test cases and test plans are complementary pillars of software quality assurance. In the Inventory Management System, **test cases** ensure each function — adding an item, updating stock, and generating reports — produces correct results for all valid, invalid, and boundary inputs. The **test plan** provides the strategic framework that organises these test cases into a managed, scheduled, and risk-aware process. Together, they reduce defects, improve system reliability, and provide documented evidence of quality — all critical requirements for any production-grade inventory system.